

Model for atomic polarize ability

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In electrostatics we model atoms as a sphere (radius a) with a uniform charge density.

$$\vec{E}_{cloud}(x) = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{q}{a^3} \vec{x} & \text{if } r = |x| < a \\ \frac{q}{4\pi\epsilon_0 |x|^2} & \text{if } r > a \end{cases} \quad (1)$$

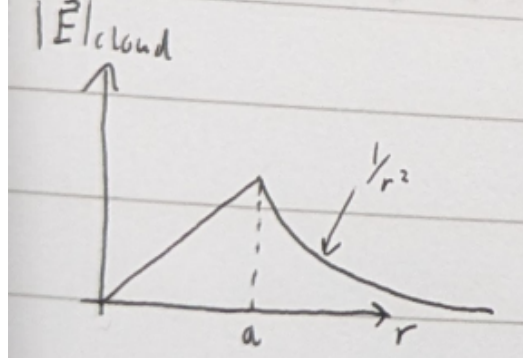


Figure 1: 4.1.png

Now if we apply an electric field. The centre of the distribution is displaced wrt the nucleus until

$$\vec{E}_{ext} + \vec{E}_{cloud}(x_0) = 0 \implies x_0 = -\frac{4\pi\epsilon_0 a^3}{q} E_{ext} \quad (2)$$

This displacement generates an electric dipole moment $\vec{d} = q\vec{x}_0 = 4\pi\epsilon_0 a^3 \vec{E}$. From now on we will call $\vec{E}_{ext} = \vec{E}$. and we call $\alpha = 4\pi\epsilon_0 a^3$. where α is the atomic polarizability.

If we consider a material made up of a lattice of atoms. The polarization $\vec{p}$

$$\vec{p} = \frac{\text{averaged } \vec{d}}{\text{unit volume}} \quad (3)$$

$$\vec{p} = n \cdot \vec{d} = n\alpha \vec{E} \quad (4)$$

where n is the density of the matter.

In general $\vec{P}$ can be a complicated function of $\vec{E}$ in linear dielectrics we have $\vec{P} = \epsilon_0 \chi_e \vec{E}$ with $\chi_e > 0$. In this simple model this is realised with $\epsilon_0 \chi_e = n\alpha$.

Electric potential of a dipole moment.

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{d} \cdot \vec{x}}{r^3}, r = |\vec{x}| \quad (5)$$

If we integrate over all dipoles *need IBP*

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int_V d^3\vec{x}' \frac{\vec{p}(\vec{x}') \cdot (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} \quad (6)$$

$$\vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} : \frac{\partial}{\partial x'_k} \left(\sum_{i=1}^3 (x_i - x'_i)^2 \right)^{-\frac{1}{2}} = -\frac{1}{2} \left(\sum_i (x_i - x'_i)^2 \right)^{-\frac{3}{2}} (-2)(x_k - x'_k) \quad (7)$$

$$= \frac{x_k - x'_k}{|\vec{x} - \vec{x}'|^3} \quad (8)$$

$$\Rightarrow \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} = \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} \quad (9)$$

$$\Rightarrow \phi(x) = \frac{1}{4\pi\epsilon_0} \int_V d^3\vec{x}' \vec{P}(\vec{x}') \cdot \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} \quad (10)$$

$$\text{IBP} \Rightarrow \frac{1}{4\pi\epsilon_0} \left[\int_{\partial V} d\vec{S} \cdot \frac{\vec{p}(\vec{x}')}{|\vec{x} - \vec{x}'|} - \int_V d^3\vec{x}' \frac{\vec{\nabla}' \vec{P}(\vec{x}')}{|\vec{x} - \vec{x}'|} \right] \quad (11)$$

Where the first term is the potential for the surface charge density on ∂V , $\sigma_{\text{bound}}(\vec{x}) = \beta \cdot \vec{n}$, with $\vec{n}$ being the outward normal. The second term including neg is the potential for $P_{\text{bound}}(\vec{x}) = -\vec{\nabla}' \vec{P}(\vec{x}')$.

Gauss's law $\nabla \cdot E = \frac{\rho}{\epsilon}$ always valid bro.

Electric displacement = $\vec{D}$

In matter we can split $\rho = \rho_{\text{free}} + \rho_{\text{bound}}$ where ρ_{free} charges move freely, ρ_{bound} is due to the polarisation.

Rewrite

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho_{\text{free}} + \rho_{\text{bound}}}{\epsilon_0} = \frac{1}{\epsilon_0} (\rho_{\text{free}} + \nabla \cdot P) \quad (12)$$

$$\vec{D} \equiv \epsilon_0 \vec{E} + \vec{P} \Rightarrow \vec{D} = \epsilon_0 \nabla \cdot E + \nabla P = \rho_{\text{free}} \epsilon \quad (13)$$

In the case of linear dielectrics:

$$\vec{P} = \epsilon_0 \chi_e \vec{E} \quad (14)$$

$$\vec{D} = \epsilon_0 (1 + \chi_e) \vec{E} = \epsilon \vec{E} = \epsilon_0 \epsilon_r \vec{E} \quad (15)$$

where we define $\epsilon_r = 1 + \chi_e$ as the relative permittivity. $\epsilon_r \approx 1$ for gasses and $\epsilon_r \approx 80$ for water.

(Magnetic Fields in Matter)

Atoms have magnetic moments caused by:

electrons with angular momentum $\rightarrow$ magnetic dipole moment,

electron spins (QM)

$$M(x) = n \langle \vec{m} \rangle \quad (16)$$

$$\langle m \rangle = \frac{1}{N} \sum_{i=1}^N m(\vec{x}_i) \quad (17)$$

Where M is magnetization. n is the 'number density of atoms' and $\langle m \rangle$ is the average magnetic moment per atom.